

CY 554: Science and Technology of Nanomaterials

Course Code	CY 554
Course Name	Science and Technology of Nanomaterials
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG
Distribution	Elective for B.Tech (all Branches), M.Sc. (Chemistry) & Ph.D.
Approval	9th Senate (May 2016)

Course Contents

- **Module I**
Fundamentals of nanoscience, surface and volume, surface energy, classification of Nanostructures [10 Lectures]
- **Module II**
Synthesis of nanomaterials, Top down, bottom up, particle stabilization in colloids, thin film deposition technologies; CVD, PVD, PLD, ALD [7 Lectures]
- **Module III**
Advanced Characterisation techniques: scanning probe microscopy, scanning electron microscopy, transmission electron microscopy [7 Lectures]
- **Module IV**
Interesting nanomaterials: metals, semiconductors, metal oxides, Fullerenes, Carbon nanotubes, graphene6Applications: Energy materials, Catalysts, sensors, display systems, nanobiotechnology, Biomimetics.[8 Lectures]
- **Module V**
Societal implications of nanotechnology: Ethical aspects, Nanotoxicology [2 Lectures]
- **Quiz I & Quiz II [2 Lectures]**

Text Books

1. C. P. Poole (Jr.) and F. J. Owens, **Introduction to Nanotechnology**, Wiley Interscience.
2. M.D. Ventra, S. Evoy, J.R. Heflin Jr. (Eds.), **Introduction to Nanoscale Science and Technology**, Kluwer Academic Publishers.

References

1. L. M. Liz-Marsan and P. V. Kamat, **Nanoscale Materials**, Kluwer Academic Publishers.
2. G. L. Hornyak, H.F. Tibbals, J. Dutta and J. J. Moore, **Introduction to Nanoscience and Nanotechnology**, CRC Press.
3. G. Cao, **Nanostructures & Nanomaterials: Synthesis, Properties & Applications**, Imperial College Press, 2004.
4. D. A. Bonnell, **Scanning Probe Microscopy and Spectroscopy: Theory, Techniques and Applications**, 2nd Edition, Wiley-VCH.
5. D. B. Williams & C. B. Carter, **Transmission Electron Microscopy: A Textbook for Materials Science**, 2nd Edition, Springer 2009.